

MIA MORRISON



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Website

PROFESSIONAL SUMMARY

Aerospace Engineering student with experience in robotics, mechanical design, systems integration, and technical project development. Recognized statewide leader and two-time Florida Youth of the Year with a proven record of developing programs, leading teams, organizing large-scale initiatives, and communicating complex ideas to diverse audiences. Passionate about applying systems thinking, engineering design, and leadership to solve real-world challenges.

EDUCATION

EMBRY-RIDDLE AERONAUTICAL UNIVERSITY (ERAU) | B.S. Aerospace Engineering | GPA: 3.33

Est. May 2027

ENGINEERING WORK EXPERIENCE

DOCKING ROBOTICS | MECHANICAL ENGINEERING INTERN | SolidWorks

May 2025 – Sept 2025

- Designed and 3D-printed a modular drone landing dock in SolidWorks using a six-panel interlocking system optimized for strength, precision, and rapid assembly.
- Conducted material selection, stress analysis, and torque calculations to improve mechanical performance; currently integrating servo-based rotational actuation for automated movement.

TECHNICAL PROJECTS

PATHFINDER AODR-01 AUTONOMOUS ROVER | Arduino, Onshape, 3D Printing, PCBWay Sponsorship

June 2026 – Present

- Independently designed and built an autonomous obstacle-detection rover using an Arduino Nano, ultrasonic sensor, DC motors, LCD display, and custom 3D-printed chassis, with self-taught Arduino programming and embedded systems.
- Selected for a PCBWay manufacturing sponsorship; redesigned prototype currently in professional production.**

SIMULINK 3D ROCKET LAUNCH SIMULATION | MATLAB, Simulink

July 2026

- Developed a 3D rocket flight simulation modeling variable-mass dynamics, thrust, drag, gravity, and translational motion in MATLAB and Simulink.
- Implemented 3-2-1 Euler attitude kinematics, body-to-inertial force transformations, and a custom MATLAB trajectory animation with real-time trajectory and flight telemetry updates.

PYTHON Satellite Trajectory Analysis Tool | Python: Tkinter, NumPy, Matplotlib, PyAstronomy, SGP4

July 2026

- Developed an interactive desktop application for satellite trajectory analysis and orbit visualization using Python.
- Implemented real-time 3D orbit visualization through adjustable Keplerian orbital elements and generated altitude profiles from TLE data using the SGP4 propagator. Integrated data visualization, GUI development, and CSV export.

MATLAB ACTIVE RECALL STUDY SYSTEM

April 2026

- Developed a MATLAB study guide program that presents questions, evaluates responses, provides immediate feedback, and explains solutions to reinforce understanding of course material.

CATIA SCOOTER DESIGN

Sept 2023 – Dec 2023

- Reverse-engineered and modeled a complete electric scooter assembly using CATIA.
 - Created detailed engineering drawings, technical documentation, and assembly models.
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Software Skills: Simulink, MATLAB, Python, Arduino IDE, SolidWorks, Onshape, CATIA, Microsoft 365

Leadership Skills: Public Speaking, Project Management, Team Leadership, Youth Mentorship, Community Engagement

LEADERSHIP & COMMUNITY EXPERIENCE

NATIONAL PUBLIC SPEAKER | YOUTH ADVOCATE (Boys & Girls Clubs of America)

June 2022- Present

- Selected as the 2022 & 2023 Florida Youth of the Year; Advocated for mental health awareness and educational resources.
- Created and led a college readiness program mentoring students through applications, scholarships, and essays; delivered keynote presentations at national conferences, including NASCAR Business.